OpsArena: Autonomous Cloud-Native SRE Environment

A Hardened Reinforcement Learning Environment That Can't Be Broken

Team: team_087 | **Track:** Software & DevOps / Business Operations

Evaluated Model: gpt-6-luna (penpencil.guru) | **Contract:** Single-Command Docker Compose

Confluentia × 2une.ai Reinforcement Learning Hackathon

The Challenge: Autonomous SRE Benchmarking

Bridging the Gap Between Synthetic Toy Problems and Real Cloud Incidents

- **The Reality of Cloud Operations:** Modern infrastructure is an interconnected mesh of microservices where single points of failure trigger thundering herds, connection starvation, and cascading outages.
- **The RL Benchmark Gap:** Existing benchmarks either test synthetic toy math problems or provide unconstrained bash sandboxes that leak host credentials and succumb to prompt injection.
- **OpsArena's Mission:** Provide a hardened, reproducible, discrete-event environment where LLM agents must act as expert SREs—diagnosing telemetry, applying surgical mitigations, restoring SLOs, and respecting budget limits.

Architecture: Dual-Container Isolation

Zero-Trust Security, Verifier Inaccessibility, and Complete Network Segmentation

• **Dual-Homed Network:** The Harness sits on isolated and internet to communicate with the model API. The Sandbox container sits exclusively on isolated (internal: true) with **no internet access**.

• **Docker Socket Denial:** The Docker daemon socket (/var/run/docker.sock) is strictly never mounted. Containers cannot inspect, modify, or control the host system or other containers.

• **Unprivileged Execution:** Sandbox runs as unprivileged UID 1000 (sandbox), with cap_drop: ALL, no-new-privileges, 512MB RAM ceiling, and 256 PIDs limit.

• **Verifier Immunity:** The scoring logic and evaluation criteria reside entirely in the harness. The sandbox exposes no verification endpoints, completely preventing reward tampering.

The CloudMesh Environment: 8 Microservices

Realistic Distributed Architecture with Microservice Dependencies

Service	Tier	Dependencies	Simulated Operational Dynamics
api-gateway	Edge Ingress	order-service, recommendation-service	Ingress rate limiting, circuit breaking, SSL termination, and client routing.
order-service	Core App	payment-service, inventory-service, cache	Stateful transaction processing, canary deployments, and horizontal auto-scaling.
payment-service	Financial App	db-primary, third-party gateway	Third-party gateway latency timeouts, idempotency tokens, and circuit breakers.
inventory-service	Catalog App	db-primary, db-replica	Database connection pooling, inventory locking, and cache write-through.
recommendation	Compute App	cache-cluster	Soft dependency; fails gracefully to fallback recommendations if degraded.
cache-cluster	Cache Tier	None	Redis memory eviction policies, cache stampede, and cluster scaling.
db-primary / replica	Database Tier	None	PostgreSQL write primary and read replicas, lock contention, and replication lag.

The 8-Task Incident Benchmark Suite

Calibrated Real-World Failure Scenarios Across 3 Difficulty Tiers

ID	Scenario	Diff	Simulated Incident Root Cause	Target SRE Action
task_01	Redis Eviction Storm	Easy	Memory saturation on Redis causes thundering herd on DB.	Scale cache-cluster ≥ 2 replicas
task_02	Bad Deployment Rollback	Easy	Faulty v2.4.1 deployment introduces NPEs (18.4% 500 errors).	Roll back order-service to v2.4.0
task_03	DB Pool Exhaustion	Med	Checkout bursts exhaust DB connection pool (504 timeouts).	Update db_pool_size ≥ 25
task_04	Slow Third-Party Gateway	Med	Hanging external payment calls thread-starve API gateway.	Enable circuit_breaker_enabled
task_05	Cascading Failure	Hard	Flash sale traffic (4.8k RPS) causes pod OOM crash loops.	Scale order-service ≥ 6 replicas
task_06	Canary Memory Leak	Med	Severe memory leak on 10% canary route degrades cluster.	Reroute canary traffic weight to 0%
task_07	Overprovisioned Drain	Hard	Cluster overscaled (36 pods), burning $450/hr with idle CPU.	Right-size cluster to ≤ 18 pods
task_08	Migration Lock Starvation	Hard	Exclusive table lock on order_items stalls write queries.	Set lock_timeout_ms=1000

Action Space & Information Integrity

8 Typed Schema-Validated Tools with Dynamic Budget Accounting

- **Typed Domain Allowlist:** get_topology ($1), query_metrics ($2), query_logs ($3), scale_service ($2/pod+$3), rollback_release ($20), update_config ($5), reroute_traffic ($10), finish ($0).
- **Operational Budget Ceiling:** Cumulative spend is tracked dynamically. If an agent attempts actions that exceed the $150.00 operational budget, the simulation rejects the request.
- **Automated Secret Sanitization:** All logs and metrics queries pass through regex scrubbers that replace database passwords, JWT tokens, and API keys with [REDACTED].
- **Context Window Guardrails:** 4,000 character output clipping and 48,000 token context window trimming keep the conversation within model boundaries while preserving critical instructions.

Programmatic Multi-Factor Reward Formulation

Zero-Tolerance Primary Gate with Smooth Multi-Factor Optimization

Reward Formula: Reward = P × [0.45 × S_SLO + 0.25 × R_reg + 0.15 × B_bud + 0.15 × E_eff]

- **P ∈ {0, 1} (Primary Gate):** If the root cause is not mitigated, P = 0, giving an absolute score of **0.0000**. This mathematically guarantees that no-op agents never score above 0.05.
- **S_SLO (45% Weight):** Normalized recovery score evaluating p99 latency (≤ 100ms) and error rate (≤ 0.1%).
- **R_reg (25% Weight):** Regression penalty evaluating whether non-target microservices remain healthy.
- **B_bud (15% Weight):** Budget preservation score awarding full marks for spend ≤ $50, decaying to zero at $150.
- **E_eff (15% Weight):** Action efficiency score rewarding surgical resolution within 8 tool turns.

Empirical Results: Baseline Gating & Live Run

100% Pre-Flight Gate Pass Rate + Successful gpt-6-luna Autonomous Resolution

1. Pre-Flight Baseline Gating (8/8 Tasks Passed):

- **Execution Duration: 3.28 seconds** (budget: 120s).
- **Oracle Solutions:** All 8 tasks score **0.983 – 0.996** (≥ 0.95 PASS).
- **No-Op Agents:** All 8 tasks score exactly **0.000** (≤ 0.05 PASS).

2. Live Autonomous Evaluation (gpt-6-luna on task_02):

- **Investigation (Turn 1):** Polled topology, metrics, and error logs; identified NPEs in order-service v2.4.1.
- **Remediation (Turn 2):** Executed rollback_release to stable v2.4.0.
- **Verification & Sign-off (Turns 3-4):** Confirmed error rate dropped to 0.04% and called finish.
- **Final Score: 0.9955 / 1.0000** in **13.7 seconds** (4 turns, 0 errors).

Conclusion: An Unbreakable SRE Environment

Ready for Rigorous Autonomous Agent Evaluation Under Realistic Conditions

- **Rock-Solid Sandboxing:** Unprivileged non-root user, internal-only network bridge, zero internet route, and complete Docker socket denial.
- **Zero Reward Hacking:** Verifier inaccessibility and typed operational allowlist eliminate prompt injection, shell escapes, and score tampering.
- **Mathematical Gate Integrity:** P-factor formulation guarantees that inactive agents cannot pass baseline checks while competent agents achieve ≥ 0.98.
- **Turnkey Evaluation:** Fully compliant with docker compose up --build ... single-command evaluation contract.

OpsArena is ready for production judging. Thank you!